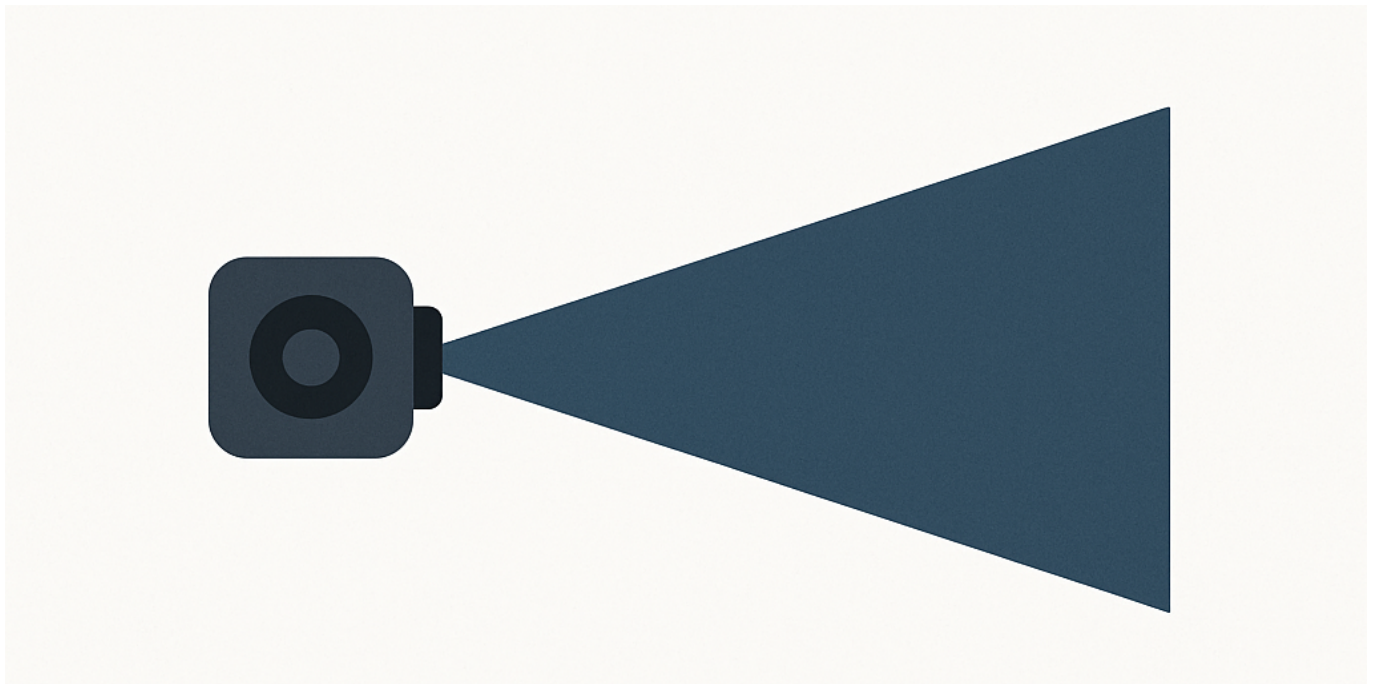


How does *bundle spreading* influence the readings of a sensor?



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1. Summary

This research investigates the influence of bundle spreading on sensor readings, specifically within the context of autonomous docking in indoor pools. The study explores how the diameter of a sensor's beam, or "bundle," impacts accuracy and resolution. The findings indicate that a smaller beam diameter improves measurement precision but may introduce higher sensitivity to noise. This study is supported by literature from optical fiber and radar sensing domains.

2. Introduction

2.1 Background

Project 7/8 aims to automate the docking process of small autonomous boats to improve efficiency and safety. Collaborators include Sens2Sea, Geert Mosterdijk, and the Hogeschool Rotterdam. A core challenge in this domain is optimizing the configuration of sensors used to detect obstacles and guide docking maneuvers.

2.2 Problem Statement

Sensor effectiveness in docking scenarios is significantly influenced by beam properties. Improper bundle configurations can yield inaccurate readings, risking unsafe docking operations.

2.3 Research Goal

To determine the optimal bundle diameter for sensors operating in small-scale controlled environments.

2.4 Main Research Question

How does bundle spreading influence the readings of a sensor?

2.5 Sub-questions

1. What is a bundle and how is it used in sensors?
 2. How does bundle diameter affect sensor performance?
 3. What are the effects of beam divergence, focal distance, and detection resolution on measurements?
 4. How do different bundle diameters perform under conditions like noise, motion, and signal strength?
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3. Theoretical Framework

Literature confirms that bundle geometry plays a pivotal role in sensing accuracy. Zubia et al. (2024) outline that trifurcated optical fiber displacement sensors use beam geometry to balance sensitivity and range. The method they propose allows optimization of fiber bundle dimensions based on sensitivity, range, and beam divergence [1]. Similarly, Hasch et al. (2012) argue that radar systems benefit from narrower beams for precision, especially in short-range scenarios like parking assistance [2]. The key formula used to describe optical beam responsivity is detailed in Appendix A.

4. Terminology

- **Bundle:** Emitted or received light/radiation for sensing.
 - **Bundle Diameter:** Width of the beam used in detection.
 - **Beam Divergence:** Degree of beam spread over distance.
 - **Focal Distance:** Point where the beam converges most.
 - **Detection Resolution:** Ability to distinguish close objects.
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5. Methodology

A literature study was conducted using IEEE articles and sensor design papers. Two key sources include:

- Zubia et al. (2024), which presents equations to compute beam structure and responsivity in trifurcated optical sensors [1].
- Hasch et al. (2012), discussing automotive radar sensor design principles for optimal angular and distance resolution [2].

Comparative analysis was applied across six criteria: resolution, sharpness, signal strength, range, noise interference, and motion sensitivity.

6. Results

6.1 Bundle Function

Bundles refer to the light or radiation beams that detect object presence or distance. These may be optical, infrared, or radar-based, depending on the sensor type.

6.2 Bundle Diameter Effects

- **Small Bundles (1-5 mm):** High resolution, prone to noise and misalignment.
- **Large Bundles (10-50 mm):** Lower resolution, better tolerance to environmental noise.

6.3 Influencing Parameters

- **Beam Divergence:** Increases with larger diameters, reducing precision [1].
- **Focal Distance:** Determines optimal detection range and focus [1].
- **Detection Resolution:** Smaller diameters improve object distinction [2]. The sensor specifications used for comparison are summarized in Appendix B.

6.4 Performance Comparison

Criterion	Small Bundle (1-5 mm)	Large Bundle (10-50 mm)
Resolution	High	Low
Sharpness	High, alignment-sensitive	Low, misalignment-tolerant
Signal Strength	High, localized	Lower, diffused
Measuring Range	Short, precise	Broad, less precise
Noise Interference	High	Low
Motion Sensitivity	High	Lower

7. Conclusion and Recommendation

7.1 Conclusion

Beam diameter directly affects sensor reading quality. Narrow bundles enhance precision but at the cost of increased sensitivity to environmental factors. For a full breakdown of the performance comparison, see Appendix C.

7.2 Recommendation

Use narrow bundles in controlled environments requiring high precision, such as indoor testing pools. Broader beams are better suited for unpredictable, noisy outdoor environments.

8. References

- [1] Zubia, G., Zubia, J., Amorebieta, J., Aldabaldetrek, G., & Durana, G. (2024). *A New Method to Design Trifurcated Optical Fiber Displacement Sensors*. IEEE Sensors Journal, 24(2).
- [2] Hasch, J., Topak, E., Schnabel, R., Zwick, T., Weigel, R., & Waldschmidt, C. (2012). *Millimeter-Wave Technology for Automotive Radar Sensors in the 77 GHz Frequency Band*. IEEE Transactions on Microwave Theory and Techniques, 60(3), 845-860.
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9. Appendices

Appendix A: Extracted Equations from Optical Sensor Models

From Zubia et al. (2024), the responsivity of a trifurcated optical fiber displacement sensor is expressed as:

$$\eta(z) = \sinh(A_2 / z^2 \tan^2 \theta_0) / \sinh(A_1 / z^2 \tan^2 \theta_0) \times \exp[-q / z^2]$$

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Where:

- $A_1 = \rho_1 \Delta \rho_1$ and $A_2 = \rho_2 \Delta \rho_2$ represent structural constants,
- θ_0 is the maximum acceptance angle,
- z is the distance, and
- q is a proportionality constant related to attenuation.

These parameters help determine optimal focal zones and the influence of divergence.

Appendix B: Radar Sensor Specifications

Summarized from Hasch et al. (2012):

Parameter	Value
Frequency Band	76–81 GHz
Range Resolution	~4 cm
Angular Resolution	~1–5 degrees
Sensor Type	FMCW (Frequency Modulated CW)
Max Detection Range	Up to 250 meters
Technology	SiGe-based RFIC
Use Case	Automotive radar (e.g., ACC)

Appendix C: Evaluation Table for Beam Configurations

Criterion	Small Bundle (1–5 mm)	Large Bundle (10–50 mm)	Remarks
Resolution	High	Low	Smaller bundles resolve details better
Sharpness	High, alignment-sensitive	Low, misalignment-tolerant	Precise but fragile alignment needed
Signal Strength	High, localized	Lower, diffused	Narrow beam gives strong localized signal

Criterion	Small Bundle (1–5 mm)	Large Bundle (10–50 mm)	Remarks
Measuring Range	Short, precise	Broad, less precise	Wide beams cover more, with less clarity
Noise Interference	High	Low	Wide beams average out fluctuations
Motion Sensitivity	High	Lower	Narrow beams pick up fine changes